Sritan Pittala

sritan.pittala@duke.edu | (302) 287-6609 | https://www.linkedin.com/in/sritan-pittala

Education

Duke University | Durham, NC Expected Graduation: May 2028

B.S.E. in Mechanical Engineering | Materials & Aerospace Certificate *GPA: 3.96/4.00*

- Relevant Coursework: Dynamics, Mechatronics, Thermodynamics, Statics, Materials, Design I & II

Experience

Chassis Lead Engineer Aug. 2024 – Present

Duke University Motorsports (Formula SAE) *Durham, NC*

- Own chassis subsystem architecture including vehicle structure, ergonomics, and driver interface/safety systems; coordinate cross-team packaging; increased chassis subsystem design points by 27.7%
- Introduced fixture-based frame manufacturing by designing and producing 100+ precision tube fixtures cutting welding time by 50% while maintaining ±3 mm frame geometry accuracy.
- Develop MATLAB and ANSYS models to analyze chassis compliance effects on lateral load transfer, refining torsional stiffness targets and reducing frame weight 10%.
- Designed team's first SLS-printed steering wheel with modular PCB housing and TPU gaskets, reducing electronics service time from minutes to seconds.

Mechanical Engineering Intern May 2026 - Aug. 2026

Dominion Defense Systems | Rinehart Racing *Asheville, NC*

- Led mechanical design of a 200-ft anti-intrusion perimeter fence for George Washington University; used hand calculations and FEA to size structural tubing for intrusion and static loading.
- Designed and analyzed an 3,000-lb center gate with a 1,500-lb architectural arch reaching 15 ft; sized support columns and hinges for gate, arch, and intrusion loads and carried design through GWU approval.
- Released 80+ manufacturing drawings for 1,500+ components and led 2 month-long production; designed 2 universal welding fixtures for repeatable assembly.
- Designed and built 4 custom shipping pallets for completed assemblies, optimizing layouts for forklift, truck, and DOT constraints; created installation drawings, BOMs, and assembly sequences for deployment at GWU.

Mechanical Engineering Researcher Aug. 2024 – Feb. 2025

Bergin Lab | Duke University *Durham, NC*

- Engineered the full mechanical design of a soot concentration sensor, including a sliding filter-tape mechanism for air sampling and automated particle analysis.
- Developed servo-actuated lifting system and custom gasket-sealed tubing integrated with a vacuum pump and digital microscope.
- Achieved 99% cost reduction while maintaining accuracy and precision comparable to commercial instruments.

Research & Development Intern June 2023 – Aug. 2024

Van Buren Labs | University of Delaware *Newark, DE*

- Developed bioinspired UUVs for naval applications across two summer R&D internships, integrating 3D-printed structures, ballast/pump systems, waterproof housings, onboard electronics, and servo-actuated propulsion.
- Led 2024 mechanical design of a modular turtle-inspired UUV with dual-servo fins and interchangeable gasketed components for rapid species reconfiguration.

Projects

Remote Sensing | Yagi-Uda Antenna & REDOX Sensor Aug. 2022 - March 2024

- 2022-2023: Designed and built Yagi-Uda antenna that achieved the strongest directional signal strength in competition, earning back-to-back state titles.
- 2024: Built an Arduino-based redox sensor to measure solute concentration via voltage detection; placing 3rd in state competition.

Technical Skills

CAD & Simulation: SolidWorks, Fusion360, Onshape, AutoCAD, ANSYS Mechanical, MATLAB
Programming: Java, Python, Arduino (C++)
Fabrication: 3D printing, laser-cutting, water jetting, silicone molding, sheet metal bending, manual milling